Does Calibration-Based Model Selection Improve Sports Betting Returns?

NBA Work

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Abstract

Walsh and Joshi (2024) claim that choosing an NBA betting model by calibration instead of accuracy turns a −35.17% return into +34.69%. We rebuild their pipeline and test it on nine seasons instead of one. On their own betting season we get the opposite ordering. Across nine seasons, calibration wins in 3 of 9, with means of −17.3% and −9.5% and a standard deviation of 62 points. We show why: the standard error of one season's return is about ±43 points, so their gap is noise. We then test whether the model knows anything the bookmaker does not. It does not: the market has lower log loss in 9 of 9 seasons, and adding our features to the market price makes predictions worse in 8 of 9. The bookmaker's margin is 3.87%, which no measured edge covers. During this work the authors published a corrigendum correcting their returns to −9.77% and −26.78%; our independent result agrees with the correction.

1 Setup and Notation

For game i let $Y_i = 1$ if the home team wins. A model outputs $\hat{p}_i = P(Y_i = 1 \mid X_i)$. The bookmaker posts American moneylines, converted to decimal odds

$$o = \begin{cases} 1 + m/100, & m > 0 \quad \text{(underdog)} \\ 1 + 100/|m|, & m < 0 \quad \text{(favourite)} \end{cases} \tag{1}$$

so a winning \$1 stake returns o in total. The implied probability is $1/o$. The two sides do not sum to one:

$$\Pi \;=\; \frac{1}{o_h} + \frac{1}{o_a} \;=\; 1.0387, \tag{2}$$

and the excess $\Pi - 1 = 3.87\%$ is the bookmaker's margin. Removing it gives the market's probability

$$q_i \;=\; \frac{1/o_h}{\Pi}, \tag{3}$$

which is our benchmark throughout.

2 Features

Per-game differential. For team t, game g, and statistic s,

$$d^{(s)}_{t,g} \;=\; s_{t,g} - s_{\mathrm{opp}(t,g),\,g}, \tag{4}$$

how far team t outperformed its opponent that night.

Season-to-date average, lagged. For game g in season S,

$$\bar{d}^{(s)}_{t,g} \;=\; \frac{1}{g-1}\sum_{k=1}^{g-1} d^{(s)}_{t,k}, \tag{5}$$

excluding game g itself. In code this is `shift(1).expanding().mean()` within each (team, season) group, which guarantees no outcome enters its own predictor.

Home-minus-away extraction. The feature for game i is

$$x^{(s)}_i \;=\; \bar{d}^{(s)}_{h,g_h} - \bar{d}^{(s)}_{a,g_a}, \tag{6}$$

giving one row per game. With 25 statistics plus the previous-season win-percentage difference, $x_i \in \mathbb{R}^{26}$.

Derived statistics. Possessions are estimated as

$$\mathrm{POSS} = \mathrm{FGA} - \mathrm{OREB} + \mathrm{TOV} + 0.44\,\mathrm{FTA}, \tag{7}$$

and the advanced quantities follow Basketball-Reference:

$$\mathrm{eFG\%} = \frac{\mathrm{FGM} + 0.5\,\mathrm{FG3M}}{\mathrm{FGA}}, \qquad \mathrm{TS\%} = \frac{\mathrm{PTS}}{2(\mathrm{FGA} + 0.44\,\mathrm{FTA})}, \tag{8}$$

$$\mathrm{ORtg} = \frac{100\,\mathrm{PTS}}{\mathrm{POSS}}, \qquad \mathrm{DRtg} = \frac{100\,\mathrm{PTS}_{\mathrm{opp}}}{\mathrm{POSS}}, \tag{9}$$

$$\mathrm{ORB\%} = \frac{\mathrm{OREB}}{\mathrm{OREB} + \mathrm{DREB}_{\mathrm{opp}}}, \qquad \mathrm{TOV\%} = \frac{100\,\mathrm{TOV}}{\mathrm{FGA} + 0.44\,\mathrm{FTA} + \mathrm{TOV}}. \tag{10}$$

Exclusions. Each team's first 10 games per season are dropped as modelling rows but kept inside the averages in Eq. (5). After this, 14,471 games remain, of which the home team wins 58.5%.

3 Screening and Selection

Drift. Each feature is compared between training and validation seasons by the two-sample Kolmogorov–Smirnov statistic

$$D = \sup_x \big|F_{\mathrm{train}}(x) - F_{\mathrm{val}}(x)\big|, \tag{11}$$

and dropped if $p < 0.01$. One feature (offensive rebounds) is dropped; 25 remain.

Redundancy (subset A). For any pair with Spearman $|\rho_{jk}| > 0.7$, we keep whichever correlates more strongly with Y and drop the other. This leaves 9 features.

Wrapper (subsets B and C). Sequential forward selection with logistic regression, run to exhaustion, keeping the best point on the path. The calibration branch minimises classwise-ECE; the accuracy branch maximises accuracy.

Calibration metric. With $M = 20$ equal bins $B_1, \ldots, B_M$,

$$\mathrm{ECE} \;=\; \sum_{j=1}^{M} \frac{|B_j|}{n}\Big|\,\overline{y}(B_j) - \overline{\hat{p}}(B_j)\,\Big|, \tag{12}$$

where $\overline{y}(B_j)$ is the realised win rate in bin j and $\overline{\hat{p}}(B_j)$ the mean prediction there. To stop a model scoring well by predicting the base rate everywhere, we set ECE = 1 whenever fewer than 80% of bins are occupied.

4 Betting Rule

A wager is placed when the model's probability exceeds the implied probability,

$$\hat{p}_i > \frac{1}{o_h} \quad \text{(back home)}, \qquad 1 - \hat{p}_i > \frac{1}{o_a} \quad \text{(otherwise back away)}. \tag{13}$$

Stakes are either flat ($100) or eighth-Kelly,

$$k = \frac{\hat{p}\, b - (1 - \hat{p})}{b}, \qquad b = o - 1, \qquad \text{stake} = \tfrac{1}{8}\, k\, B, \tag{14}$$

with bankroll B starting at $10,000. Return on investment is $\text{ROI} = (B_{\text{end}} - B_0)/B_0$.

Break-even. A bet at decimal odds o needs a hit rate of $1/o$ to break even, so the edge required is

$$\varepsilon_{\min}(o) \;=\; \frac{1}{o} - q, \tag{15}$$

which shrinks as o grows. This single fact explains most of our results: the same edge in percentage points is worth far more on an underdog than on a favourite.

Variance. With n bets at odds o and true hit rate p, profit per $100 stake has standard deviation

$$\sigma = 100\, o \sqrt{p(1-p)}, \qquad \sigma_{\text{total}} = \sigma\sqrt{n}. \tag{16}$$

For $n = 450$, $o = 4.5$, $p = 0.28$ this gives $\sigma_{\text{total}} \approx \$4{,}286$, i.e. ± 43 ROI points on a $10,000 bankroll. A single season cannot resolve a difference smaller than this.

5 Single-Season Replication

We first reproduce the original design exactly: train on 2014–17, select on 2017–18, bet on 2018–19.

	LR	RF	MLP	SVM
Classwise-ECE (branch B)	**0.0431**	0.0568	0.0465	0.0440
Accuracy (branch C)	0.6530	0.6540	0.6549	**0.6633**

Table 1: Test-season scores. Selected: LR on calibration, SVM on accuracy.

Both land inside the original paper's reported ranges (3.23–4.39% ECE, 65.34–66.55% accuracy), so the pipeline reproduces. The betting season does not.

Model	Rule	Final	ROI	% bet	% won
Calibration (LR)	flat	$10,097	+0.98%	85.3	38.0
Calibration (LR)	Kelly	$6,500	−35.00%	85.3	38.0
Accuracy (SVM)	flat	$12,503	+**25.03**%	88.6	34.9
Accuracy (SVM)	Kelly	$4,489	−55.11%	88.6	34.9

Table 2: 2018–19 betting simulation. The ordering is reversed relative to the original study, and both Kelly variants lose.

The accuracy-driven model earns more despite being worse on both metrics it was not selected for: its ECE on the betting season is 100% (the bin constraint fires, $\sigma_{\hat{p}} = 0.142$) against 4.27% for the calibration model. A model with degenerate calibration returning +25% is the first indication that returns here are not driven by forecast quality.

6 Nine-Season Extension

We repeat the entire pipeline on a rolling origin: for each i, train on seasons $i, \ldots, i+3$ and bet on season $i+4$.

Betting season	$\lvert B\rvert$	$\lvert C\rvert$	Calibration	Accuracy
2014–15	2	7	−65.30%	−16.78%
2015–16	5	4	−17.73%	−12.31%
2016–17	4	4	+**57.42**%	+46.75%
2017–18	4	6	−100.00%	−100.00%
2018–19	2	4	+2.08%	+12.25%
2019–20	5	4	+85.58%	+100.86%
2020–21	6	6	−31.59%	−33.85%
2021–22	3	7	−83.67%	−89.28%
2022–23	3	3	−2.89%	+6.93%
Mean			−17.34%	−9.49%
Std. dev.			61.7	62.5
Calibration wins			3 of 9	

Table 3: Flat staking, rolling origin. Both means are negative and the ordering is a coin flip.

The observed spread of 61.7 points matches the ±43 predicted by Eq. (16) to within sampling error, confirming that the design generates far more noise than signal. Two seasons deserve comment. In 2017–18 both systems reach −100%: flat \$100 staking over roughly 950 bets commits \$95,000 against a \$10,000 bankroll, so ruin is reachable, a possibility the original betting rule does not address. In 2016–17 and 2019–20 both systems gain 45–100%, and Section 7 shows these are seasons in which *any* underdog strategy profited.

7 Why: Market Efficiency

7.1 The model never beats the price

Comparing $\hat{p}_i$ against q_i from Eq. (3):

	Log loss	Accuracy	ECE
Model	0.6230	64.90%	0.1554
Market	**0.5962**	**67.91**%	**0.0384**

Table 4: Pooled over nine betting seasons. The market is better on every metric; it wins on log loss in 9 of 9 seasons.

The market's own calibration error of 3.84% is lower than the best model reported in the original paper.

7.2 The features add nothing

Fit three logistic models and compare out-of-sample log loss: market logit alone, box-score features alone, and both together.

If the features carried residual information, the fused model would improve on market-only. It does not. This is the mechanical reason no betting rule built on these features can work.

Features	Log loss	AUC	Brier
Box score only	0.6230	0.696	0.2170
Market only	0.5972	0.732	0.2055
Fused	0.6003	0.728	0.2070
Raw market price	**0.5958**	**0.733**	**0.2050**

Table 5: Adding box-score features to the market price degrades log loss in 8 of 9 seasons. Nothing beats simply reading the odds.

7.3 Prices are calibrated at every level

Bucketing all games by q and comparing to realised frequency, the largest gap is 1.4 points, and every bucket returns -2% to -4% under flat staking, the margin from Eq. (2). Testing the underdog-bias hypothesis directly, with no model involved, gives $z = 0.15$: there is no systematic mispricing to exploit.

7.4 Where the returns actually came from

Decomposing profit by the price of the side backed:

Bucket	Bets	Implied	Actual	Edge	Profit
Longshot ($< 35\%$)	3,591	22.01%	23.34%	+1.33	+\$5,376
Dog (35–50%)	2,113	41.70%	41.60%	−0.10	−\$8,660
Favourite (50–65%)	1,395	57.18%	56.49%	−0.69	−\$6,679
Big favourite ($> 65\%$)	805	72.18%	73.54%	+1.36	−\$1,446

Table 6: Every profitable season is carried by the longshot bucket; every losing season by longshot misses.

The two edge-positive buckets illustrate Eq. (14). Both show roughly +1.3 points, but only the longshots profit. At $q = 0.72$ the price is about 1.24, so $0.7354 \times 1.24 = 0.985 < 1$; the required edge is 2.8 points and we have 1.4. At $q = 0.22$ the price is about 4.4, the required edge is 0.7 points, and 1.33 clears it.

The longshot edge is nonetheless not significant: over 3,591 wagers at a 23% base rate the standard error is ± 1.38 points against an observed +1.33 ($z = 1.32$). Confirming an edge of this size would require roughly 15,000 wagers, about forty NBA seasons.

7.5 Restrictions do not rescue it

Restriction	Bets	Edge	Yield
All value bets (original rule)	7,904	+0.59	−1.44%
Underdogs only	5,704	+0.80	−0.58%
Favourites only	3,030	+0.46	−3.07%
Favourites, gap > 0.08	881	+0.83	−2.35%

Table 7: No restriction reaches profitability. Removing underdogs, the intuitive safety measure, removes the better half of the strategy.

Requiring larger model–market disagreement makes matters worse, not better. Among bets where the model exceeds the market by more than 0.12, the model predicts 70.5%, the market

50.9%, and the realised rate is 51.6% — the outcome tracks the market, not the model. Large disagreement signals model error, not private information.

8 Discussion

The corrigendum. During this work the authors published a correction reporting feature-engineering errors found while releasing their code.[1] The corrected returns are −9.77% for calibration selection and −26.78% for accuracy selection, both negative, and the selected calibration model changes from SVM to MLP. The claim that Kelly staking succeeds given calibration is withdrawn. Our independent results agree in direction with the correction rather than the original.

What survives. Calibration remains the right metric for probabilistic decisions: it measures the quantity a decision rule consumes, and accuracy does not. What does not survive is the inference from calibration to profit. Against a market with a 3.87% margin that already prices every probability bucket correctly, a better-calibrated model simply agrees with the price more closely, and agreement earns nothing.

Limitations. Our odds come from one book and may be opening rather than closing quotes. The simulation charges no transaction costs and assumes unlimited liquidity at the posted price. We report a grid of 69 strategies in Section 7.5 without multiplicity correction; the apparent winners did not survive blind testing, which is consistent with selection noise.

Future work. The one avenue our evidence does not close is information the market lacks at posting time, injuries, rest and confirmed lineups, evaluated by closing-line value rather than realised ROI, since the latter needs forty seasons to resolve an edge of the size available here.

[1] `https://doi.org/10.1016/j.mlwa.2025.100627`